Sourabh Joshi

+91-8105513470 | joshisourabh1000@gmail.com | linkedin.com/in/sourabh-joshi-60a268208 | github.com/8Sourabh8

Summary

Design verification engineer trained at Maven Silicon, now pursuing an M.Tech at NIT Karnataka. Hands-on in digital design and verification: Verilog RTL, SystemVerilog, UVM, assertion-based verification (SVA) and functional coverage, with Perl/Python scripting on Unix/Linux. Seeking an ASIC digital design or formal verification role.

Education

M.Tech, Communication Engineering and Networks – NIT Karnataka, Surathkal — Aug 2026 – Present
Advanced VLSI Design and Verification Course – Maven Silicon, Bengaluru — Jul 2024 – May 2026
B.E., Electronics and Communication – KLS Gogte Institute of Technology, Belagavi – CGPA: 7.40 — 2021 – 2024
Diploma, Electronics and Communication – KLS's S. V. Potdar Polytechnic, Belagavi – 80% — 2018 – 2021
SSLC (KSEEB) – St Paul's High School, Belagavi – 76% — 2018

Design and Verification Projects

VLSI Design and Verification Projects — Jul 2024 – May 2026

Maven Silicon – SystemVerilog, UVM, Synopsys VCS, QuestaSim, SpyGlass — *Bengaluru*

- **1x3 Router:** Verilog RTL (finite state machine, synchroniser, register, 3 FIFOs) linted with SpyGlass; UVM testbench with source/destination agents, virtual sequencer, scoreboard and cross coverage; 100% functional coverage.
- **AHB-to-APB Bridge:** UVM environment with AHB master and APB slave agents, single/burst sequences and monitors.
- **UART:** UVM loopback testbench: TX-to-RX data scoreboard, baud-rate setup, FIFO-full and corner-pattern tests.
- **Pipelined Dual-Port RAM and MOD-10 Counter:** self-checking SystemVerilog testbenches with constrained-random stimulus and assertions for resets and corner cases; 100% functional and code coverage.

RTL Phase Controller for RIS-Assisted mmWave MIMO — Oct 2026

M.Tech term project, NITK – Verilog, Vivado, Artix-7 FPGA, MATLAB

- Designed a Verilog controller (codebook ROM, FSM, shift register, latch); 18/18 self-checking tests passed.
- Artix-7: 157 LUTs, 147 FFs, timing met at 100 MHz; beam update in 1.31 μs, 100x faster than channel change.
- MATLAB: reproduced 1-/2-bit RIS quantisation loss (3.6 / 0.85 dB); coverage behind a building raised from 8.8% to 100%.

28 GHz Inset-Fed Microstrip Patch Antenna — Oct 2026

M.Tech project, NITK – COMSOL Multiphysics

- 4.2 x 3.5 mm patch on RT/duroid 5880; studied feed and fabrication tolerance; bandwidth widened from 640 to 920 MHz.

Research Projects and Presentations

Electrochemical Detection of Herbicide Molinate Using ZnO Nanoparticles — Sep 2022 – Jan 2023

Nano-Club research, KLS Gogte Institute of Technology

- Modified a carbon paste electrode with ZnO nanoparticles; cyclic voltammetry gave a detection limit of 10 nM.
- Applied to spiked soil and water samples with 91.7%–99.5% recovery.

Frequency-Based Electro-Culture for Chemical-Free Farming — Dec 2023 – Apr 2024

UG major project, KLS Gogte Institute of Technology – Raspberry Pi, antenna design

- Built a copper antenna that channels atmospheric ions into the soil; tested on turnip plants across 5 growth measures (water content, chlorophyll/SPAD, height, biomass, leaf area) with improved growth and no chemical fertilisers.
- Won **1st Prize, Project Expo 2024** (ECE, innovative project category).

Conference Presentations

- **Poster**, International Conference on Electrochemistry in Industry, Health and Environment (EIHE-2023), BARC Mumbai, Feb 2023.
- **Oral presentation**, International Conference on Advances in Materials and Chemistry (ASMC-2023), Mar 2023.

Internship

Intern – Amsa Embedded Solutions — Jan 2024 – Mar 2024

Embedded systems – NodeMCU ESP8266, UART, I2C, DipTrace — *Hubli*

- IoT health monitor streaming heart rate, SpO2, temperature, humidity to ThingSpeak; designed PCBs, debugged UART/I2C.

Technical Skills

Verification: SystemVerilog, UVM, SystemVerilog Assertions (SVA), constrained-random verification, functional and code coverage
Digital Design: Verilog RTL, finite state machines, FIFOs, pipelining, linting and synthesis, FPGA implementation
Scripting: Perl, Python, C, MATLAB, Makefile, Unix/Linux shell
EDA Tools: Synopsys VCS, SpyGlass, Design Compiler, QuestaSim, Xilinx ISE, Vivado, Cadence Virtuoso, LTSpice
Other: AMBA AHB/APB, UART, I2C protocols; COMSOL, Altium, DipTrace

Achievements and Certifications

- 1st Prize, Best Innovative Project – Maven Silicon; 1st Prize, Project Expo 2024 – KLS Gogte Institute of Technology.
- NPTEL: The Joy of Computing Using Python (Jul 2023); Linguaskill Business B2 – Cambridge English (Jun 2022).
- Placement Class Representative; Art Co-ordinator, Association of Electronics Engineers – KLS Gogte Institute of Technology.